



Yahya Ibrahim

PH.D. · AI / ML ENGINEER & RESEARCHER

CONTACT

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RESEARCH IMPACT

11

PAPERS

163

CITATIONS

6

H-INDEX

5

I10-INDEX

TECHNICAL SKILLS

Programming

Python MATLAB C++

DL Frameworks

PyTorch TensorFlow Keras

Lightning

Model Architectures

Transformer Autoregressive

CNN GAN

3D & Vision

NeRF 3D Gaussian Splatting

OpenCV Open3D PCL

Tools & Infra

ROS2 Docker Git W&B

HPC/Linux LaTeX

EDUCATION

PhD · PPKE ITK, Budapest

2017 – 2023

MSc · Tishreen Univ., Syria

2013 – 2016

BSc · Tishreen Univ., Syria

2009 – 2013

LANGUAGES

Arabic (Native) English (Fluent)

Hungarian (Basic)

PROFESSIONAL SUMMARY

AI/ML engineer and researcher with a PhD (PPKE ITK, 2023) and six years building production-grade deep learning systems at HUN-REN SZTAKI's Machine Perception Laboratory. Proven track record delivering end-to-end solutions across 3D scene reconstruction (NeRF, 3D Gaussian Splatting, point cloud completion), generative image synthesis (GANs, inpainting), and satellite/remote sensing pipelines — with live demos, open-source code, and 163 citations. Experienced with the full ML lifecycle: data, architecture design, training at scale on HPC, and deployment. Seeking **AI/ML Engineer** or **Research Scientist** roles in industry.

EXPERIENCE

AI/ML Researcher — Machine Perception Laboratory

2017 – Present

HUN-REN SZTAKI, Budapest, Hungary

- Designed and trained **3D point cloud completion networks** (PyTorch) for autonomous driving — deployed as open-source tool with 11 citations in Elsevier IVC.
- Built **GAN-based blind inpainting** pipeline for heritage masonry restoration; 39-citation MDPI paper and live interactive demo at imgproc.mplab.sztaki.hu.
- Developed **satellite image segmentation** system (SplitClass network, Sentinel-1/2) for real-time inland flood monitoring — HuggingFace demo + ICCVW 2025.
- Conducted **EEG signal classification** using deep recurrent CNNs for BCI applications; most-cited work (57 citations, IEEE BCI 2020).
- Developing **AI perception for drones and underwater LiDAR** using ROS2 within the EU Proactif project.
- International research visit**, Nanyang Technological University (NTU), Singapore (Nov 2025) — collaboration on geospatial AI and generative multimedia with S-Lab and CoNiC Lab.

KEY PROJECTS & DEMOS

3D Point Cloud Completion — MVPCC-Net

PyTorch · Open3D · Autonomous Driving

Multi-view deep learning for completing partial LiDAR scans. End-to-end pipeline from raw sensor data to completed 3D mesh. Published in Elsevier IVC (2023).

github.com/sztaki-geocomp/Multi-view-3d-completion

Satellite Flood Detection — SplitClass

Sentinel-1/2 · Segmentation · HuggingFace

Real-time inland excess water monitoring from multispectral satellite imagery. Live demo deployed on HuggingFace Spaces. ICCVW 2025 (IEEE/CVF).

waterdetection.github.io · huggingface.co/spaces/Yahyasztaki/water_flooding_1

Masonry Wall Inpainting & Restoration

GAN · CNN · Live Demo

GAN-based blind inpainting for heritage wall damage detection and reconstruction. Live interactive web demo. 39 citations (MDPI Remote Sensing, 2020).

imgproc.mplab.sztaki.hu · mplab.sztaki.hu/geocomp/masonryWallAnalysis

SELECTED PUBLICATIONS · Full list at scholar.google.com/citations?user=1ShbTvIAAAA

Inland Excess Water (IEW) Monitoring Using Sentinel-1/2: A SplitClass Segmentation and Temporal Gap-Filling Approach

Y. Ibrahim, M. Belényesi, C. Liu, M. Richter-Cserey, M. Simon, T. Sziranyi, C. Benedek

ICCVW — IEEE/CVF, SEA Workshop · 2025

MVPCC-Net: Multi-View Based Point Cloud Completion Network for MLS Data

Y. Ibrahim, C. Benedek

Image and Vision Computing — Elsevier · 2023 · **11 citations**

Multi-View Based 3D Point Cloud Completion Algorithm for Vehicles in Autonomous Driving

Y. Ibrahim, B. Nagy, C. Benedek

ICPR — IEEE · 2022 · **7 citations**

Multi-Class Classification of Motor Imagery EEG Signals Using Deep Recurrent CNN

W. Fadel, C. Kollod, M. Wahdow, **Y. Ibrahim**, I. Ulbert

8th International BCI Meeting — IEEE · 2020 · **57 citations**

Deep Learning-Based Masonry Wall Image Analysis

Y. Ibrahim, B. Nagy, C. Benedek

Remote Sensing — MDPI · 2020 · **39 citations**

PROFESSIONAL SERVICE

Peer reviewer · 4 Elsevier Q1 journals · 9 reviews · ORCID verified

Engineering Applications of AI · 2025

Automation in Construction · 2024

Measurement · 2024